



NAME:SAMARTH CHAND THAKURI

SAP ID:500119430

BATCH:B1 CCVT

```
1.package Experiment8;
import java.io.*;
import java.util.Scanner;
public class First {

    public static void readFile(String fileName) throws
FileNotFoundException, IOException {
        BufferedReader reader = new BufferedReader(new
FileReader(fileName));

        System.out.println("File contents:");
        String line;
        while ((line = reader.readLine()) != null) {
            System.out.println(line);
        }
        reader.close();
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the file name: ");
        String fileName = scanner.nextLine();

        try {
            readFile(fileName);
        } catch (FileNotFoundException e) {
            System.out.println("Error: File not found. Please
check the file name and try again.");
        } catch (IOException e) {
            System.out.println("Error: An I/O error occurred
while reading the file.");
        } finally {
            System.out.println("File operation attempted.");
        }
    }
}
```

```
    }

    scanner.close();
}
}
```

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\javac.exe' /d C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab\src /s C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab\src\*.java
Enter the file name: student.txt
File contents:
Name: Samarth, Roll Number: 45, Grade: 79
Name: Samarth, Roll Number: 34, Grade: 23
File operation attempted.
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> 
```

2. `package Experiment8;`

```
import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;
```

```
public class Second {
    public static void main(String[] args) {
```

```
Scanner scanner = new Scanner(System.in);

System.out.print("Enter student name: ");
String name = scanner.nextLine();

System.out.print("Enter roll number: ");
String rollNumber = scanner.nextLine();

System.out.print("Enter grade: ");
String grade = scanner.nextLine();

String studentData = "Name: " + name + ", Roll Number: "
+ rollNumber + ", Grade: " + grade + "\n";

try (FileWriter writer = new FileWriter("student.txt",
true)) { // 'true' enables appending
    writer.write(studentData);
    System.out.println("Student record added
successfully.");
} catch (IOException e) {
    System.out.println("An error occurred while writing
to the file: " + e.getMessage());
}

scanner.close();
}
```

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-X
ing\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java\jdt_ws\OOPS-lab-_16c6cd2b\bin' '
Enter student name: samarth
Enter roll number: 34
Enter grade: o
Student record added successfully.
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> |
```

3. `package Experiment8;`

```
import java.io.File;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.IOException;

public class Third {
    public static void main(String[] args) {
        File file = new File("student.txt");

        try (FileReader fileReader = new FileReader(file)) {
            int character;
            while ((character = fileReader.read()) != -1) {
                System.out.print((char) character);
            }
        } catch (FileNotFoundException e) {
```

```
        System.out.println("Error: The file 'student.txt'
was not found.");
    } catch (IOException e) {
        System.out.println("Error reading the file: " +
e.getMessage());
    }
}
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\
ing\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java\jdt_ws\O
Name: Samarth, Roll Number: 45, Grade: 79
Name: Samarth, Roll Number: 34, Grade: 23
Name: samarth, Roll Number: 34, Grade: o
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab->
```

4. `package Experiment8;`

```
import java.io.File;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.IOException;
import java.util.Scanner;

public class Fourth{
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the file name: ");
        String fileName = scanner.nextLine();

        File file = new File(fileName);

        if (!file.exists() || !file.isFile()) {
            System.out.println("Error: The file '" + fileName +
"'" does not exist or is not a valid file.");
            return;
        }

        int charCount = 0;
        int wordCount = 0;

        try (FileReader fileReader = new FileReader(file);
            Scanner fileScanner = new Scanner(file)) {

            while (fileScanner.hasNext()) {
                String word = fileScanner.next();
                wordCount++;
                charCount += word.length();
            }

            System.out.println("Total words: " + wordCount);
```

```
        System.out.println("Total characters (excluding  
whitespace): " + charCount);  
  
    } catch (FileNotFoundException e) {  
        System.out.println("Error: The file '" + fileName +  
"'" was not found.");  
    } catch (IOException e) {  
        System.out.println("Error reading the file: " +  
e.getMessage());  
    } finally {  
        scanner.close();  
    }  
}  
}
```

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\java.exe'  
ing\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java\jdt_ws\OOPS-lab-_16c6cd2b\bin  
Enter the file name: student.txt  
Total words: 21  
Total characters (excluding whitespace): 104  
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> |
```


5.package Experiment8;

```
import java.io.*;

class Person implements Serializable {
    private static final long serialVersionUID = 1L; // Ensure
version control during serialization
    private String name;
    private int age;

    public Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    public String getName() {
        return name;
    }

    public int getAge() {
        return age;
    }

    @Override
    public String toString() {
        return "Person{name='" + name + "', age=" + age + "}";
    }
}

public class Fifth{
    public static void main(String[] args) {
        String fileName = "person.txt";
        Person person = new Person("John Doe", 25);

        // Serialize the Person object
    }
}
```

```
        try (ObjectOutputStream oos = new ObjectOutputStream(new
FileOutputStream(fileName))) {
            oos.writeObject(person);
            System.out.println("Person object serialized
successfully.");
        } catch (IOException e) {
            System.out.println("Error serializing object: " +
e.getMessage());
        }

        // Deserialize the Person object
        try (ObjectInputStream ois = new ObjectInputStream(new
FileInputStream(fileName))) {
            Person deserializedPerson = (Person)
ois.readObject();
            System.out.println("Deserialized Person: " +
deserializedPerson);
        } catch (IOException | ClassNotFoundException e) {
            System.out.println("Error deserializing object: " +
e.getMessage());
        }
    }
}
```

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab-> & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-XX:+S
ing\Code\User\workspaceStorage\abb8d221f0bb4c8398f033aa5831e6bc\redhat.java\jdt_ws\OOPS-lab-_16c6cd2b\bin' 'Expe
Person object serialized successfully.
Deserialized Person: Person{name='John Doe', age=25}
PS C:\Users\samar\OneDrive\ドキュメント\GitHub\OOPS-lab->
```